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Tech Skills

C/C++, Python, PyTorch, TensorFlow

LLMs/SLMs, VLMs, VLAs

Architecture, Alignment (SFT, RLHF),
Parallelism (Triton, DDP, FSDP),
Inference (Quantization, Speculative
Decoding), Agentic Pipelines

Reinforcement Learning

Online, Offline, Imitation / Curriculum /
Adversarial Learning, League Training

Computer Vision

Object Detection, Semantic
Segmentation, Behavior Classification

Modeling and Simulation

Multi-Body Dynamics, Atmospheric &
Space Flight Dynamics, Robotics
Systems, Sensor Fusion

Infrastructure & Tools

vLLM, Docker, Kubernetes, CI/CD, Git,
AWS/GCP

Soft Skills

Strategic Thinking

Aligning development with real-world
impact

**First-Principles and Out-of-the-Box
Thinking**

Reasoning from fundamentals,
approaching problems with
unconventional perspectives

Creative Problem Solving

Tackling open-ended,
high-uncertainty challenges

Cross-Functional Collaboration

Bridging engineering, research, and
domain experts

Self-Directed Leadership

Driving projects from concept to
deployment

Technical Communication

Explaining complex AI to diverse
audiences

Alessandro Palmas

Senior AI/ML Research Engineer

Senior AI/ML Research Engineer with 15+ years of experience building intelligent systems that combine foundation models, reinforcement learning, high-fidelity simulation, and multimodal AI. Experienced in turning foundational research into scalable, production-grade solutions across aerospace, defense, robotics, and gaming. Expertise spans agent design, training pipelines, model alignment, evaluation benchmarks, and AI-native systems, with a track record of bridging deep research and large-scale engineering to tackle real-world challenges.

Experience

2025/10 - present

LawZero - Montreal • Permanent Full-time

Senior AI/ML Research Engineer

I'm part of the core research and engineering team at LawZero, a new non-profit AI research lab led by Yoshua Bengio. The lab's mission is to advance foundation models that are truthful, transparent, and safe-by-design, systems that understand the world rather than merely act in it.

- Contributing to early research and engineering efforts on reasoning, alignment, and interpretability.
- Exploring scalable implementations and training frameworks for safe foundation models.
- Designing and implementing benchmarking frameworks to evaluate model reasoning and safety.
- Collaborating with cross-disciplinary teams to shape the lab's technical direction and infrastructure.

2022/12 - 2025/10

Ubisoft La Forge - Montreal • Permanent Full-time

Senior AI/ML Research Engineer

I led the development of next-generation AI/ML systems for AAA games, with a focus on Deep Reinforcement Learning and simulation-driven agent training. My work spanned research, tooling, and deployment, delivering scalable learning pipelines and integrating AI behaviors into live game environments.

- Built and deployed Deep Reinforcement Learning systems for integration into AAA game environments.
- Designed scalable curriculum and multi-agent RL algorithms including League Training.
- Worked on Online, Imitation, Offline, and Hierarchical RL, explored Multimodal and Foundation Model Agents.
- Integrated AI models into proprietary engines for NPC behavior, testing, and live simulations.

Education

MSc by Research - Post-Grad Space Engineering
University Of Glasgow (2013)

MSc Aerospace Engineering
Milan Polytechnic (2010)

MSc Aerospace Engineering
Turin Polytechnic (2010)

BSc Aerospace Engineering
Turin Polytechnic (2008)

Publications

[Linkedin](#) | [Google Scholar](#)

Language

Italian (Native)

English (Professional)

Spanish (Intermediate)

French (Intermediate)

Interests & Passions

Building AI-based Tech

Open Source

History of Mathematics

2018/02 - present

Artificial Twin • Freelance

AI/ML Consultant

Providing consulting and development services in advanced AI/ML systems across industries, with a focus on Reinforcement Learning, LLMs/VLMs/VLAs, computer vision, and physics-based modeling.

- Designed and delivered intelligent agents for decision-making and control through custom RL pipelines in simulation and real-world settings, leveraging curriculum learning, imitation learning, and hybrid modeling.
- Explored the integration of transformer-based LLMs and emerging Vision-Language-Action Models into end-user applications for reasoning, retrieval, and control.
- Delivered computer vision systems for detection, segmentation, and semantic understanding in industrial and robotics domains, optimized for deployment on constrained compute environments.
- Developed and maintained proprietary libraries supporting ML (RL, CV, LLM), physics-based simulation, and spatial intelligence workflows.

2020/11 - 2024/12

DIAMBRA (Acquired in Dec 2024) • Part-time

Founder & Principal AI/ML Engineer

Founded and led DIAMBRA, a cloud-native platform for competitive multi-agent RL. Enabled researchers and practitioners to train, benchmark, and deploy agents in complex environments.

- Built a distributed platform for tournament-style benchmarking with support for cloud inference, model submission, and automated evaluation.
- Developed cross-platform environments based on competitive fighting games. Supported workflows for curriculum design, human-in-the-loop training, imitation learning, and agent-vs-agent/human competition.
- Released tools as open-source Dockerized packages, fostering a community of RL researchers and ensuring reproducibility and real-world applicability.
- Led DIAMBRA to acquisition, validating the product's technical foundation and commercial relevance.

2017/07 - 2022/11

Nurjana Technologies • Permanent Full-time

Principal Research Engineer

Led development of AI-driven autonomous systems, perception, and simulation for aerospace and defense applications. Delivered real-world solutions across aerial ISR, satellite collision avoidance, and robotics, integrating ML, embedded systems, and physical modeling.

- Deployed real-time computer vision models (object detection, segmentation, tracking, activity recognition) on edge devices (e.g., NVIDIA Jetson) for drones operating in GNSS-denied zones.
- Built hardware-in-the-loop pipelines with Unreal Engine to validate vision-based navigation and target detection in photorealistic environments.
- Developed custom flight control, vision-based landing, and obstacle avoidance for autonomous aerial platforms. Integrated vision, control, and simulation into deployable systems.
- Designed software for orbit propagation, attitude dynamics, conjunction analysis, maneuver planning, and sensor tasking. Contributed to operational space situational awareness systems.